ADDITIVE FUNCTORS ARE EXACT ON HOMOTOPY CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3663-3673, following the statement at 3647-3661.

Editorial note. No omissions: the transfer of splittings is checked termwise and the four bounded variants are the same argument.

Allowed standard prerequisites (editorial).

- Termwise split sequences and their associated morphism. definition-splitses (same chapter, cited at ~3667). Supplies splittings (s^n, pi^n) and delta.

Source reference. Original definition 014I: sources/stacks-derived/pinned-original-source.tex, source lines 3036–3036. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$, $\mathcal{B}$ be additive categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. The induced functors*

$$\begin{aligned} F : K(\mathcal{A}) &\longrightarrow K(\mathcal{B}) \\ F : K^+(\mathcal{A}) &\longrightarrow K^+(\mathcal{B}) \\ F : K^-(\mathcal{A}) &\longrightarrow K^-(\mathcal{B}) \\ F : K^b(\mathcal{A}) &\longrightarrow K^b(\mathcal{B}) \end{aligned}$$

are exact functors of triangulated categories.

Proof. Suppose $A^\bullet \to B^\bullet \to C^\bullet$ is a termwise split sequence of complexes of $\mathcal{A}$ with splittings (s^n, π^n) and associated morphism $\delta : C^\bullet \to A^\bullet[1]$, see Definition 014I. Then $F(A^\bullet) \to F(B^\bullet) \to F(C^\bullet)$ is a termwise split sequence of complexes with splittings $(F(s^n), F(\pi^n))$ and associated morphism $F(\delta) : F(C^\bullet) \to F(A^\bullet)[1]$. Thus F transforms distinguished triangles into distinguished triangles. □

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